

The toughness conjecture was long expected to hold with t as small as $t = 2$. This was disproved after many years, but the general conjecture remains open. See the exercises for how the conjecture ties in with the results given in the remainder of this chapter.

It may come as a surprise to learn that hamiltonicity is also related to the four colour problem. As we noted in Chapter 6.6, the four colour theorem is equivalent to the non-existence of a planar snark, i.e. to the assertion that every bridgeless planar cubic graph has a 4-flow. It is easily checked that ‘bridgeless’ can be replaced with ‘3-connected’ in this assertion, and that every hamiltonian graph has a 4-flow (Ex. 16, Ch. 6). For a proof of the four colour theorem, therefore, it would suffice to show that every 3-connected planar cubic graph has a Hamilton cycle!

Unfortunately, this is not the case: the first counterexample was found by Tutte in 1946. Ten years later, Tutte proved the following deep theorem as a best possible weakening:

Theorem 10.1.4. (Tutte 1956)

Every 4-connected planar graph has a Hamilton cycle.

Although, at first glance, it appears that the study of Hamilton cycles is a part of graph theory that cannot possibly extend to infinite graphs, there is a fascinating conjecture that does just that. Recall that a *circle* in an infinite graph G is a homeomorphic copy of the unit circle S^1 in the topological space $|G|$ formed by G and its ends (see Chapter 8.6). A *Hamilton circle* of G is a circle that contains every vertex of G .

Hamilton
circle

Conjecture. (Bruhn 2003)

Every locally finite 4-connected planar graph has a Hamilton circle.

10.2 Hamilton cycles and degree sequences

Historically, Dirac’s theorem formed the point of departure for the discovery of a series of weaker and weaker degree conditions, all sufficient for hamiltonicity. The development culminated in a single theorem that encompasses all the earlier results: the theorem we shall prove in this section.

If G is a graph with n vertices and degrees $d_1 \leq \dots \leq d_n$, then the n -tuple $(d_1, \dots, d_n)$ is called the *degree sequence* of G . Note that this sequence is unique, even though G has several vertex enumerations giving rise to its degree sequence. Let us call an arbitrary integer sequence $(a_1, \dots, a_n)$ *hamiltonian* if every graph with n vertices and a degree sequence pointwise greater than $(a_1, \dots, a_n)$ is hamiltonian. (A sequence $(d_1, \dots, d_n)$ is *pointwise greater* than $(a_1, \dots, a_n)$ if $d_i \geq a_i$ for all i .)

degree
sequence

hamiltonian
sequence

pointwise
greater

The following theorem characterizes all hamiltonian sequences:

Theorem 10.2.1. (Chvátal 1972)

An integer sequence $(a_1, \dots, a_n)$ such that $0 \leq a_1 \leq \dots \leq a_n < n$ and $n \geq 3$ is hamiltonian if and only if the following holds for every $i < n/2$:

$$a_i \leq i \Rightarrow a_{n-i} \geq n-i.$$

$(a_1, \dots, a_n)$ *Proof.* Let $(a_1, \dots, a_n)$ be an arbitrary integer sequence such that $0 \leq a_1 \leq \dots \leq a_n < n$ and $n \geq 3$. We first assume that this sequence satisfies the condition of the theorem and prove that it is hamiltonian.

Suppose not. Then there exists a graph whose degree sequence $(d_1, \dots, d_n)$ satisfies

$$d_i \geq a_i \quad \text{for all } i \quad (1)$$

$G = (V, E)$ but which has no Hamilton cycle. Let $G = (V, E)$ be such a graph, chosen with the maximum number of edges.

By (1), our assumptions for $(a_1, \dots, a_n)$ transfer to the degree sequence $(d_1, \dots, d_n)$ of G ; thus,

$$d_i \leq i \Rightarrow d_{n-i} \geq n-i \quad \text{for all } i < n/2. \quad (2)$$

x, y Let x, y be distinct and non-adjacent vertices in G , with $d(x) \leq d(y)$ and $d(x) + d(y)$ as large as possible. One easily checks that the degree sequence of $G + xy$ is pointwise greater than $(d_1, \dots, d_n)$, and hence than $(a_1, \dots, a_n)$. Hence, by the maximality of G , the new edge xy lies on a Hamilton cycle H of $G + xy$. Then $H - xy$ is a Hamilton path $x_1, \dots, x_n$ in G , with $x_1 = x$ and $x_n = y$ say.

As in the proof of Dirac's theorem, we now consider the index sets

$$I := \{i \mid xx_{i+1} \in E\} \quad \text{and} \quad J := \{j \mid x_jy \in E\}.$$

Then $I \cup J \subseteq \{1, \dots, n-1\}$, and $I \cap J = \emptyset$ because G has no Hamilton cycle. Hence

$$d(x) + d(y) = |I| + |J| < n, \quad (3)$$

h so $h := d(x) < n/2$ by the choice of x .

Since $x_iy \notin E$ for all $i \in I$, all these x_i were candidates for the choice of x (together with y). Our choice of $\{x, y\}$ with $d(x) + d(y)$ maximum thus implies that $d(x_i) \leq d(x)$ for all $i \in I$. Hence G has at least $|I| = h$ vertices of degree at most h , so $d_h \leq h$. By (2), this implies that $d_{n-h} \geq n-h$, i.e. the $h+1$ vertices with the degrees $d_{n-h}, \dots, d_n$ all have degree at least $n-h$. Since $d(x) = h$, one of these vertices, z say, is not adjacent to x . Since

$$d(x) + d(z) \geq h + (n-h) = n,$$

this contradicts the choice of x and y by (3).

Let us now show that, conversely, for every sequence $(a_1, \dots, a_n)$ as in the theorem, but with

$$a_h \leq h \quad \text{and} \quad a_{n-h} \leq n-h-1$$

for some $h < n/2$, there exists a graph that has a pointwise greater degree sequence than $(a_1, \dots, a_n)$ but no Hamilton cycle. As the sequence

$$(\underbrace{h, \dots, h}_h \text{ times}, \underbrace{n-h-1, \dots, n-h-1}_{n-2h \text{ times}}, \underbrace{n-1, \dots, n-1}_h \text{ times})$$

is pointwise greater than $(a_1, \dots, a_n)$, it suffices to find a graph with this degree sequence that has no Hamilton cycle.

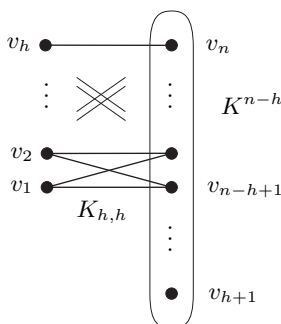


Fig. 10.2.1. Any cycle containing $v_1, \dots, v_h$ misses v_{h+1}

Figure 10.2.1 shows such a graph G , with vertices $v_1, \dots, v_n$ and the edge set

$$\{v_i v_j \mid i, j > h\} \cup \{v_i v_j \mid i \leq h; j > n-h\};$$

it is the union of a K^{n-h} on the vertices $v_{h+1}, \dots, v_n$ and a $K_{h,h}$ with partition sets $\{v_1, \dots, v_h\}$ and $\{v_{n-h+1}, \dots, v_n\}$. Deleting these latter h vertices leaves more than h components. Hence G is not 1-tough, and thus not hamiltonian. $\square$

By applying Theorem 10.2.1 to graphs of the form $G * K^1$, one can easily prove the following adaptation of the theorem to Hamilton paths. Let an integer sequence be called *path-hamiltonian* if every graph with a pointwise greater degree sequence has a Hamilton path.

Corollary 10.2.2. *An integer sequence $(a_1, \dots, a_n)$ such that $n \geq 2$ and $0 \leq a_1 \leq \dots \leq a_n < n$ is path-hamiltonian if and only if every $i \leq n/2$ is such that $a_i < i \Rightarrow a_{n+1-i} \geq n-i$. $\square$*

10.3 Hamilton cycles in the square of a graph

 G^d

Given a graph G and a positive integer d , we denote by G^d the graph on $V(G)$ in which two vertices are adjacent if and only if they have distance at most d in G . Clearly, $G = G^1 \subseteq G^2 \subseteq \dots$. Our goal in this section is to prove the following fundamental result:

Theorem 10.3.1. (Fleischner 1974)

If G is a 2-connected graph, then G^2 has a Hamilton cycle.

The proof of Theorem 10.3.1 will go roughly as follows. We start by finding a cycle C in G . Using induction, we shall cover the remaining vertices by C -paths in G^2 . The first and last edges of those paths will be edges of G , like those of C . By deleting some of these edges and doubling others, we turn the union of C and all the C -paths into a multigraph with even degrees, and find an Euler tour in it. This Euler tour W will pass some vertices more than once, but all edges in such multiple passes will be edges of G . For all but one of the passes through a given vertex we can therefore try to replace its two G -edges by an edge of G^2 (Fig. 10.3.1), hoping to turn our Euler tour into a Hamilton cycle of G^2 . The main difficulty will be to ensure that these lifts of passes are compatible, i.e., that we do not attempt to lift an edge at both its ends.

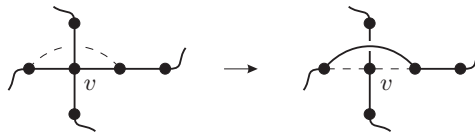


Fig. 10.3.1. Reducing the degree of v in W by lifting a pass

Lemma 10.3.2. *For every 2-connected graph G and $x \in V(G)$, there is a cycle $C \subseteq G$ that contains x as well as a vertex $y \neq x$ with $N_G(y) \subseteq V(C)$.*

Proof. If G has a Hamilton cycle, there is nothing more to show. If not, let $C' \subseteq G$ be any cycle containing x ; such a cycle exists, since G is 2-connected. Let D be a component of $G - C'$. Assume that C' and D are chosen so that $|D|$ is minimum. Since G is 2-connected, D has at least two neighbours on C' . Then C' contains a path P between two such neighbours u and v , whose interior $\overset{\circ}{P}$ does not contain x and has no neighbour in D (Fig. 10.3.2).

Replacing P in C' by a u - v path through D , we obtain a cycle C that contains x and a vertex $y \in D$. If y had a neighbour z in $G - C$, then z would lie in a component $D' \subsetneq D$ of $G - C$, contradicting the choice of C' and D . Hence all the neighbours of y lie on C , and C satisfies the assertion of the lemma. $\square$